

Guided Capstone Project Report: Recommendations for Big Mountain Resort

June 6, 2026

To the Board of Directors and Management Committee of Big Mountain Resort

Executive Summary – We analyzed 330 peer ski resorts in the United States for the results of our study presented to the Board of Directors and Management Committee of Big Mountain Resort (BMR). Our data science methodology is based on supervised predictive modeling, and more specifically, regression analysis. Our findings support the competitive advantages and unique attributes that BMR maintains over its competitors and options for pricing, venue updates, and operational enhancements. Using our modeling expertise, we determined that there are five components that largely explain consumer's willingness to pay premium ticket pricing for access to US-based resorts. Before we share those attributes and how we arrived at this level of selective focus, we will share our data science analysis techniques –their strengths and some cautions about exclusive use of these factors as the basis for changes to your pricing strategy.

Data Set – As noted, our sample population is based on 330 ski resorts in the United States (35 states total), with 24 specific attributes, and 3 resort identifiers (resort name, region, and state). We segmented these attributes into five performance categories: elevation and terrain scale, lift infrastructure, snow conditions and related capacity, operating season and resort history, and ticket pricing. The purpose of our segmentation is to allow for commonality of the 24 separate attributes, thus, giving the Board and Management the ability to consider how BMR compares with its peers in like characteristics that either differentiate the resort based on hard-limitations, such as mountain height, or those that could be addresses through operational changes, including adding additional chair lifts. The data set is summarized per table “Comparison of Big Mountain Resort Versus All Resorts in Sample Population.”

Data Science Methods Employed – We employed the following data science techniques in pursuit of determining the comparative strengths of BMR and those factors that serve as the defining attributes that drive ski lift ticket prices.

- **Principal Component Analysis (PCA)** – Dimensionality-reduction technique used to transform a dataset with many correlated variables into a smaller set of new variables called principal components.
- **Correlation Analysis** – A measure of the strength and direction of the relationship between a dependent variable and one or more independent variables.
- **Cross Validation Score** – Estimates how well the model is likely to perform on unseen data by testing it across multiple slices of the training data.
- **Random Forest Regression** – A means of predicting a number by averaging the predictions from many decision trees.
- **Scenario Analysis** – Method of testing the outcome of a prediction based upon changes to a set of specified attributes.
- **Analysis of Data Deficiency** – Concern for the limitations of data analysis and related predictive value, based upon missing data and/or sampling.

Data Set Analysis – BMR's attributes were compared with the peer-group using the mean, standard deviation, and z-score, organized within the aforementioned five-segment framework. We paid particular attention to your relative strengths and weakness using the z-score, which is denoted in the table

(Data Set Analysis – Continued)

“Comparison of Big Mountain Resort Versus All Resorts in Sample Population” using either green or red highlights. We note high positive differentiation in attributes included in three segments (elevation and terrain scale, lift infrastructure, snow conditions and related capacity), with negative differentiation in attributes in the other two segments (operating season and resort history, and ticket pricing). Instances of negative or positive differentiation was discarded for certain attributes that have low value in explaining the statistical significance of pricing power, including attributes like availability of a tram, and high-capacity chairlift arrangements (sixes and eights).

Data Set Strengths Compared to Correlation Analysis – From our correlation analysis, we note five variables that appear most positively correlated with adult weekend ticket prices, and BMR’s relative strength in these categories, based on z-scores:

Rank of Five Highest Variables/Attributes to Adult Weekend Ticket Prices & BMR's Z-Score				
Rank	Variable/Attribute	Correlation	BMR's Z-Score	Comparative Value
1	Runs	.7569	1.24	> 1 Stan. Dev. Above Peers
2	Fast Quads	.7314	0.90	~ 1 Stan. Dev. Above Peers
3	Vertical Drop	.7133	1.20	> 1 Stan. Dev. Above Peers
4	Snow Making (acres)	.6958	1.64	> 1 Stan. Dev. Above Peers
5	Total Chair Lifts	.6544	0.99	~ 1 Stan. Dev. Above Peers

The data indicates that those attributes that have the highest correlation to adult weekend ticket prices are categories that BMR enjoys comparative strength when compared to the peer-group. This provides BMR with pricing optionality, given other changes to operations and venue.

Modeling Results – The model helps reinforce our analysis and recommendations to the management team by identifying resort attributes most strongly associated with ticket-price power. Based on the train/test split, cross-validation scoring, and random forest regressor feature-importance results, the strongest predictive factors are the number of fast quad lifts, total ski runs, snow-making acreage/coverage, and the mountain’s vertical drop. We caution that the model’s predictive value does not necessarily prove causation, and, thus, conclusions on changes to pricing must be made in coordination with other considered changes in resort management, operations, and ski terrain.

Cross Validation and Analysis of Data Deficiency – Based on the data analysis technique of cross validation (CV), which estimates how well the model is likely to perform on unseen data by testing it across multiple slices of the training data, that further performance in the model is unlikely given variables in excess of eight. This is a statistical observation and cannot be empirically verified. Further, though such can be considered a valid stand-alone consideration, the following are data deficiencies:

- Assessability of BMR, or how easy is travel to the resort? Is it serviced by an airport? If so, how close is it?
- Given BMR is an independently owned and operated ski resort does it accept passes from other large mountain companies or ownership groups, such as the Epic, Ikon, Indy, or Power passes? If not, why not?
- Are there other cultural or seasonal resort events sponsored by Big Mountain? Extreme skiing competition? Concerts? Gaper Day celebration?
- Overall operating costs and profit or loss of the resort operations for the prior five years. Cost breakout by operations and cost centers for the past five years.

Summary and Executive Recommendations – We conclude our analysis for considered changes to ticket pricing based on the modeling techniques employed; the expected adult weekend lift-ticket price for the resort is \$95.87, compared with its current price of \$81.00. This \$14.87 pricing gap suggests that the resort may have some flexibility to increase its lift-ticket price. However, the model's mean absolute error is \$10.39, indicating that the magnitude of the potential increase should be evaluated cautiously. The following changes, or continuing operations, would help to lift ticket pricing:

1. We do not recommend making changes to the current number of ski runs. The correlation of lost revenue due to the shuttering of five or more runs suggests that such a change would be deleterious to the financial performance of the resort.
2. Increase the vertical drop by adding a run to a point 150 feet lower down but requiring the installation of an additional chair lift to bring skiers back up, without additional snow making coverage. This would be included as a recommendation to the management committee. Due to current tax law, costs for installing the additional chair lift could be immediately expensed for tax purposes. This significantly increases the cash flow return on investment and lessens the payback period of such a capex project.
3. Same as number 2, but adding 2 acres of snow making cover. This would be included as a recommendation to the management committee. Due to current tax law, costs for installing additional snow-making equipment could be immediately expensed for tax purposes. This significantly increases the cash flow return on investment and lessens the payback period of such a capex project.
4. Increase the longest run by 0.2 mile to boast 3.5 miles length, requiring additional snow making coverage of 4 acres. This would be included as a recommendation to the management committee. Due to current tax law, costs for installing additional snow-making equipment could be immediately expensed for tax purposes. This significantly increases the cash flow return on investment and lessens the payback period of such a capex project.
5. Add cultural and/or seasonal resort events, including extreme skiing competition, concerts, and other fun attractions such as a Gaper Day celebration.

Comparison of Big Mountain Resort Versus All Resorts in Sample Population						
Segment	Database Field	BMR Actual	All Resorts Mean	BMR - All Resort Mean	All Resorts Std. Dev.	Z-Score*
Elevation & Terrain Scale	Summit Elevation	6,817	4,585	2,232	3,739	0.60
	Vertical Drop	2,353	1,212	1,141	947	1.20
	Base Elevation	4,464	3,371	1,093	3,121	0.35
	Total Runs	105	48	57	46	1.24
	Terrain Parks	4.0	2.8	1.2	2.0	0.60
	Longest Run (mile)	3.3	1.43	1.87	1.15	1.63
	Skiable Terrain (acres)	3,000	656	2,344	1,094	2.14
Lift Infrastructure	Trams	0	0.17	-0.17	0.56	-0.30
	Fast Eights	0	0.01	-0.01	0.08	-0.13
	Fast Sixes	0	0.19	-0.19	0.65	-0.29
	Fast Quad	3	1.01	1.99	2.2	0.90
	Quad	2	0.93	1.07	1.31	0.82
	Tripple	6	1.49	4.51	1.6	2.82
	Double	0	1.84	-1.84	1.81	-1.02
	Surface	3	2.62	0.38	2.06	0.18
	Total Chairs	14	8.25	5.75	5.8	0.99
Snow Conditions & Related Capacity	Snow Making (acres)	600	173	427	261	1.64
	Average Snowfall	333	185	148	136	1.09
	Night Skiing (acres)	600	98	502	99	5.07
Operating Season & Resort History	Days Open Last Year	123	115	8	35	0.23
	Projected Days Open	123	120	3	31	0.10
	Year Open	72	64	8	110	0.07
Ticket Pricing	Adult Weekday Ticket	\$81	\$58	\$23	\$26	0.88
	Adult Weekend Ticket	\$81	\$64	\$17	\$25	0.68
Resort Identification & Geo Info	Name of Resort	BMR	N/A	N/A	N/A	N/A
	Region	Montana	N/A	N/A	N/A	N/A
	State	Montana	N/A	N/A	N/A	N/A

*Z-Score – Calculated per ((BMR Actual – (minus) All Resorts Mean) / All Resorts Std. Dev.), with:
0.00 – Indicating equal to all resorts mean / +1.00 – Indicating one Stan. Dev. above mean / -1.00 – Indicating one Stan. Dev. below mean